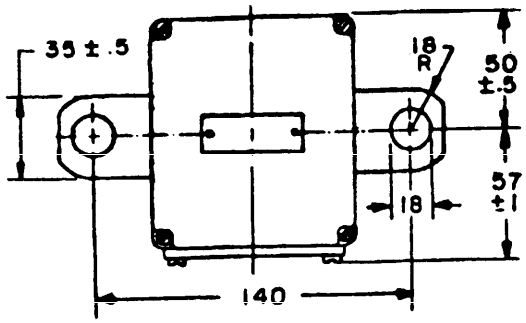
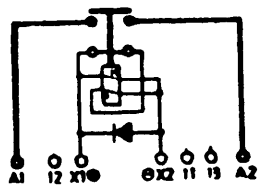
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 1000 AMPERES, SPST, N.O., D.B., TYPE II, NON-HERMETICALLY SEALED (AND MOUNTING BRACKET), METRIC

The specification sheet is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

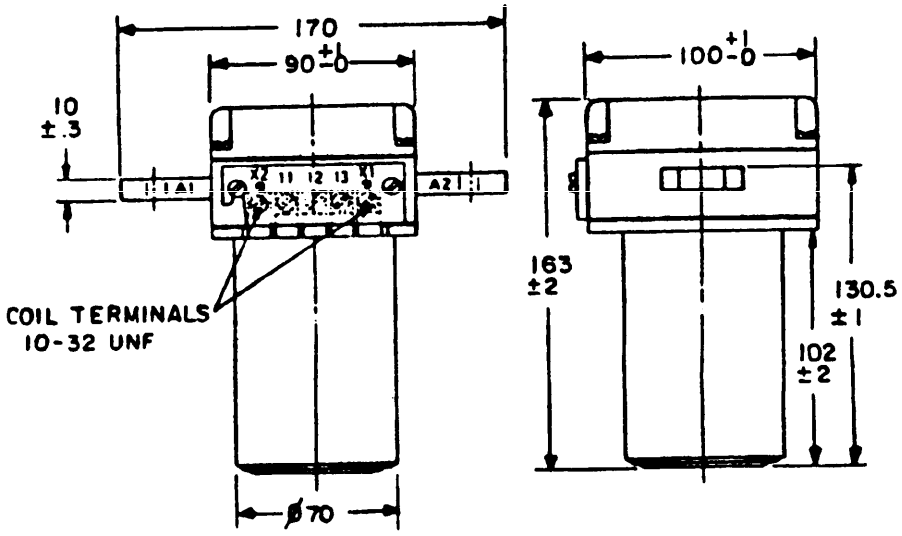
The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.





CIRCUIT:

P = PULL IN COIL
H = HOLDING COIL



NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±1.57 mm (.062 in).

MM	INCHES	MM	INCHES
0.3	.012	70	2.75
0.5	.02	90	3.54
1	.04	100	3.93
2	.08	102	4.01
10	.39	130	5.11
18	.7	140	5.51
35	1.37	163	6.41
50	1.96	170	6.69
57	2.24		

FIGURE 1. Outline drawing.

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REQUIREMENTS: (Relays)

Coil data:

Duty rating: Continuous.

Operating voltage: 28 V dc to 30 V dc.

Pickup voltage, maximum: 18 V dc over the temperature range.

Dropout voltage: 5 +0, -4 V dc, over the temperature range.

Hold voltage: 10 +0, -4 V dc.

Hold current, maximum: 2 amperes.

Resistance, pull in coil: 0.8 ohm ±20 percent.

Pull in current, maximum: 40 amperes for 30 milliseconds.

Resistance, holding coil: 18.5 ohms ±10 percent.

Time milliseconds, maximum:

Operate: 60

Release: 400

Bounce: 2.5

CONTACT DATA:

Rated contact load: See table I.

TABLE I. Rated contact load (28 V dc).

Type of load	Life operating cycles x 10^3	Load
Resistive	10	1000 amps
Inductive	5	250 amps
Motor	5	500 amps
Mechanical life	20	250 amps

PHYSICAL CHARACTERISTICS:

Dimensions and configurations: See figures 1 and 2.

Material: Steel, USAI-AISI-MP-1015.

Finish: Paint, olive.

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Weight:

Relay: 4.2 kilograms maximum (9.1 pounds).

Terminals: See figure 1.

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -40°C to +71°C.

Shock:

Energized: MIL-STD-202, test method 213, test condition A, 50 G's.

Deenergized: MIL-STD-202, test method 213, test condition A, 50 G's (except Z* axis, 30 G's).

Vibration:

Energized: MIL-STD-202, test method 204, test condition D, 20 G's.

Deenergized: MIL-STD-202, test method 204, test condition C, 10 G's.

Acceleration: 15 G's.

ELECTRICAL CHARACTERISTICS:

Insulation resistance, minimum:

Initial: 100 megohms.

After life or environmental: 50 megohms.

Dielectric strength

1 Minute	Test points
1050 Volts	Contacts to case
1050 Volts	Contacts to coil
1050 Volts	Coil to case
1050 Volts	Between open contacts

Contact drop, maximum:

Initial: 0.15 volt.

After life test: 0.175 volt.

Overload current: 4,000 amperes for 20 cycles.

Rupture current: 10,000 amperes for 20 cycles.

Part number: D6106/47- (dash number from table II).

TABLE II. Nomenclature.

-001	Figure 1	Relay
-002	Figure 2	Mounting bracket

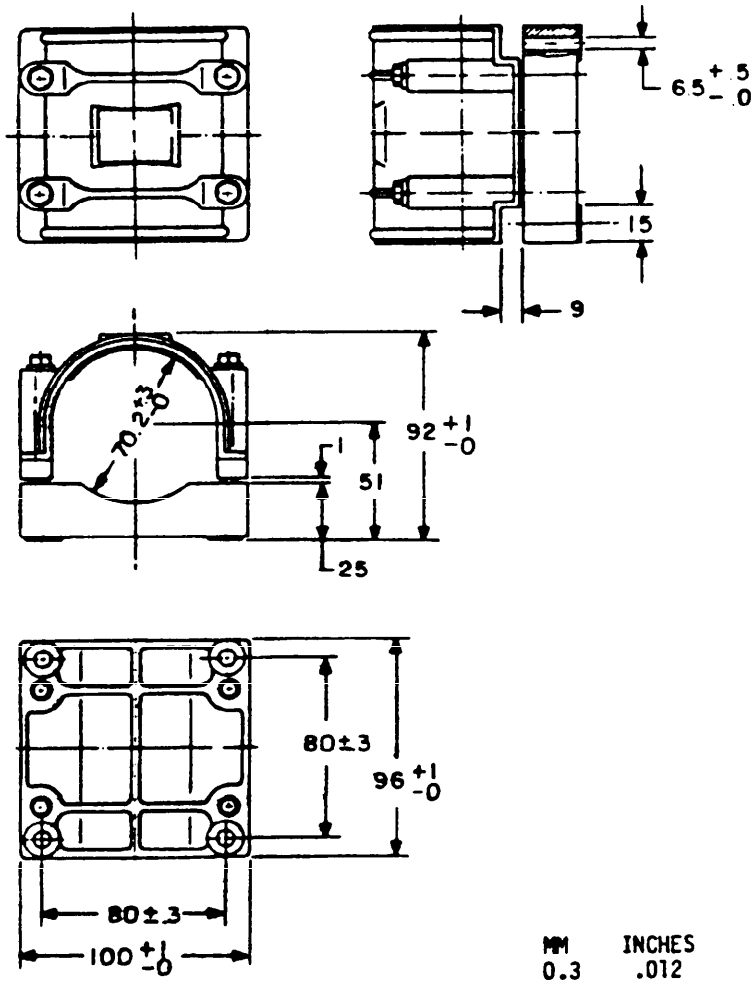
REQUIREMENTS: (Mounting bracket, see figure 2)

Material: Aluminum.

Finish: Copper surface, electroplate nickel.

Weight: 1.2 pounds.

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MM	INCHES	MM	INCHES
0.3	.012	51	2.00
0.5	.02	70.2	2.76
1	.04	80	3.15
6.5	.255	92	3.62
9	.354	96	3.77
15	.59	100	3.39
25	.98		

NOTES:
1. Dimensions are in millimeters.
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3. Unless otherwise specified, tolerance is ±1.57 mm (.062 in).

FIGURE 2. Mounting bracket

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Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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